

AlumniConnect

A Major Project Synopsis Submitted to



**Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal
Towards Partial Fulfillment for the Award of**

**Bachelor of Technology
(Computer Science and Engineering)**

**Under the Supervision of
Prof.**

Submitted By

Piyush Mishra(0827RL243D05)

Ashish Sharna(0827RL273D05)

Name 3(enroll. No)

Name 4(enroll. No)



**Department of Computer Science and Engineering
Acropolis Institute of Technology & Research, Indore**

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Instructions for Students

- The synopsis should describe the problem, proposed solution, development plan, expected outcomes, and references.
- Use clear technical language and avoid unsupported claims or copied content.
- Figures, diagrams, tables, and references should be numbered and referred to in the text wherever applicable.
- Use recent and reliable research papers, books, official documentation, and credible websites as references.
- The proposed system should be feasible within the project duration, available resources, and team size.
- All team members should have clearly defined responsibilities.

1. Introduction of the Project (150-200 words)

Introduce the background of the problem, explain why the problem is important, identify the target users, and briefly describe the proposed solution and its expected benefit

Example

Project: Smart Attendance System Using Face Recognition

Attendance management in educational institutions is often performed manually or through traditional biometric systems. Manual attendance consumes considerable classroom time and may be prone to errors and proxy attendance. The proposed Smart Attendance System uses face recognition techniques to automatically identify students and record their attendance. The system captures the student's face through a camera, compares it with registered facial data, and records attendance along with date and time. Faculty members can view attendance records through a web-based dashboard and generate reports. The proposed system aims to reduce manual effort, improve attendance accuracy, and provide an efficient method for maintaining attendance records.

2. Objective (50-100 words)

State the specific objectives of the project. What goals do you intend to achieve through this project? Objectives should begin with action-oriented terms such as **To develop, To design, To implement, To analyze, To automate, To improve, To evaluate.**

Example

Project: Smart Attendance System

Objectives:

- 1. To develop an automated attendance system using face recognition.*
- 2. To reduce the time required for manual attendance.*
- 3. To minimize errors and proxy attendance.*
- 4. To provide faculty members with attendance reports.*
- 5. To maintain attendance records in a centralized database.*

3. Scope (100 words)

Define the scope of your project. What are the boundaries and limitations of your project's functionality?

Example

Project: Smart Attendance System

In Scope

- Student registration*
- Face capture*
- Face recognition*
- Attendance recording*

- *Attendance reports*
- *Faculty dashboard*

Out of Scope

- *Integration with college ERP*
- *Mobile application*
- *Multi-campus deployment*
- *Biometric hardware integration*

4. Study of Existing System (200 words)

Include comparative case study of at least 5 existing systems with their merits and demerits. This will help in formulation of further objectives that could be addressed within the project.

Example

No	Existing System	Problems Addressed	Advantages	Disadvantages	Gap Identified	Reference
1	Google Classroom	Online learning management	Easy to use	Limited custom analytics	Limited institution-specific features	[1]
2	Moodle	LMS management	Open source	Requires configuration	UI/customization challenges	[2]
3	Microsoft Teams	Online collaboration	Collaboration tools	Requires account setup	Not specifically designed for attendance analytics	[3]
4	Existing System 4	Problem solved	Major benefits	Major limitations	Gap	[4]
5	Existing System 5	Problem solved	Major benefits	Major limitations	Gap	[5]

5. Project Description (150-250 words)

Describe the project supported by Flowcharts/ER diagram to explain the flow of the information.

5.1 Proposed System

Explain how the proposed system addresses the identified gap.

5.2 Working of the System

Explain the major steps involved.

5.3 Data Flow / Process Flow

Include flowchart or DFD.

5.4 Database Design

Include ER diagram where applicable.

6. Planning of the Project work

Describe the development approach and major phases. Include activities, milestones, dependencies, responsibilities, and a Gantt chart covering the complete project duration.

Include a Gantt chart for timeline with milestones.

Sample Example to be included

Phase	Activity	Indicative Duration
1	Problem Identification & Literature Survey	Week 1
2	Requirement Analysis	Week 2
3	System Design	Week 3
4	Database & UI Design	Week 4-5
5	Implementation	Week 6-10
6	Testing & Debugging	Week 10-11
7	Documentation(Report, Paper, Patent Doc)	Week 10-12
8	Final Demo & Presentation	Week 12

7. Features

List specific functionalities that users will be able to perform. Organize them, where applicable, into functional, user-interface, security, performance, reporting/analytics, and user-experience features.

Example: For an Online Examination System: student login, exam creation, question management, timed examination, automatic evaluation, result generation, and performance analytics.

Category	Example Features
User Interface	Login, dashboard, forms, responsive layout
Functional	Registration, search, transaction, prediction, report generation
Security	Authentication, authorization, secure password handling
Performance	Fast response, efficient database queries
Reporting & Analytics	Charts, summaries, downloadable reports
User Experience	Notifications, validation messages, accessibility

8. System architecture

Provide a high-level architecture showing how users, frontend, backend/services, database, APIs, ML models, and other components interact. Use a clear labelled diagram.

Example

Typical web application: User → Frontend → Backend/API → Database.

AI project: User → Web Interface → Backend → ML Model → Prediction → Database.

9. User Interface (UI)

Describe the user interface design. You can include Form designs, wireframes or mockups if applicable.

10. Technology Stack

List the technologies, programming languages, frameworks, and databases you'll use.

Example

Component	Technology / Tool
Frontend	HTML/CSS/JavaScript or React.js
Backend	Node.js / Django / Spring Boot
Programming Language	Python / Java / JavaScript
Database	MySQL / PostgreSQL / MongoDB
ML Framework (if applicable)	TensorFlow / PyTorch / Scikit-learn
Development Tool	VS Code / IntelliJ IDEA / Android Studio
Version Control	Git / GitHub

11. Testing Plan

Explain how the project will be tested for correctness, reliability, usability, performance, and security as applicable. Include representative test cases for important functions.

Provide tentative test cases.

Example

Test Case ID	Test Scenario	Input	Expected Output
TC01	Valid Login	Correct username/password	User successfully logged in
TC02	Invalid Login	Wrong password	Appropriate error message
TC03	Empty Input	Blank required fields	Validation message displayed
TC04	Valid Registration	Correct user details	User registered successfully
TC05	Duplicate Record	Existing ID/email	Duplicate warning displayed

12. Expected Outcome and Benefits (100-150 words)

Describe what the completed project is expected to achieve, who will benefit, measurable or observable improvements, practical applications, and possible future benefits.

Example

Project: Smart Attendance System

The proposed Smart Attendance System is expected to automate the attendance recording process and reduce the time required for manual attendance. The system will provide faculty members with accurate and easily accessible attendance records. Students will be able to view their attendance status, while faculty members can generate attendance reports. The system

may help reduce errors and proxy attendance and improve the overall efficiency of attendance management. In future, the system can be integrated with the institute ERP system and mobile applications.

13. Resources and Limitations (100 words)

1. Resources:

List the hardware, software, datasets, APIs, cloud resources, devices, or industry data required to design, develop, test, and demonstrate the project.

Example

Resource Type	Examples
Hardware	Laptop/PC, camera, sensors, mobile device, server
Software	IDE, operating system, database, frameworks
Data	Public dataset, collected dataset, APIs, industry data
Other	Internet access, cloud account, testing devices

2. Limitations:

Mention realistic constraints such as dataset size, hardware limitations, internet dependency, limited testing environment, accuracy constraints, scalability, or features intentionally excluded from the current version.

Example

The model may show lower accuracy under poor lighting or with limited training data. The initial version may be tested with a limited number of users.

14. Conclusion (100-150 words)

Conclude by briefly connecting the problem, proposed solution, major objectives, expected benefits, and future potential. Do not introduce new information or features in the conclusion.

15. References

List all sources actually referred to in the synopsis. Use a consistent citation style. Include research papers, books, official documentation, standards, datasets, and credible websites as applicable. Avoid unreliable or unattributed sources.

Source Type	Recommended Format
Research Paper	[1] A. Author and B. Author, "Title of Paper," Journal/Conference, vol., no., pp., Year.
Book	[2] A. Author, Title of Book, Edition. Publisher, Year.
Website	[3] Organization/Author, "Title of Web Page," Website Name, Year.